EXPERIMENT 1

CONFIGURATION OF NETWORK COMPONENTS

Aim

To study the following network devices in detail:

- PC
- Server
- Repeater
- Hub
- Switch
- Bridge
- Router
- Gateway
- Transmission medium

Software Used

Cisco Packet Tracer

Procedure

Step 1: Start Cisco Packet Tracer.

Step 2: Select each device type from the device box at the bottom-left (End Devices, Network Devices) and place the PC, Server, Repeater, Hub, Switch, Bridge, Router and Home Gateway in the workspace.

Step 3: Study each device using the information given below.

Node: A node is a connection point in a network that can receive, create, store or send data.

PC: A personal computer is an end device used by a person to send and receive data on the network.

Server: A server is a computer on the network that manages network resources and gives services to other computers.

Repeater (Physical Layer): A repeater receives a signal and retransmits it at higher power, so the signal can travel longer distances.

Hub (Physical Layer): A hub connects many devices so that they act as a single network segment. It is a multiport repeater and sends the data it receives to all ports.

Switch (Data Link Layer): A switch connects network segments and forwards data only to the port of the destination device using MAC addresses.

Bridge (Data Link Layer): A bridge connects two network segments and decides whether a packet should be sent to the other segment. A switch is a bridge with many ports.

Router (Network Layer): A router connects two or more networks and forwards packets between them using the address information in each packet and its routing table.

Gateway: A gateway connects networks that use different protocols. It converts the protocols so that the networks can work together.

Transmission medium: The path through which signals travel from one device to another. Guided media use cables, such as twisted pair and coaxial cable. Unguided media do not use cables, such as air.

Execution Screenshot

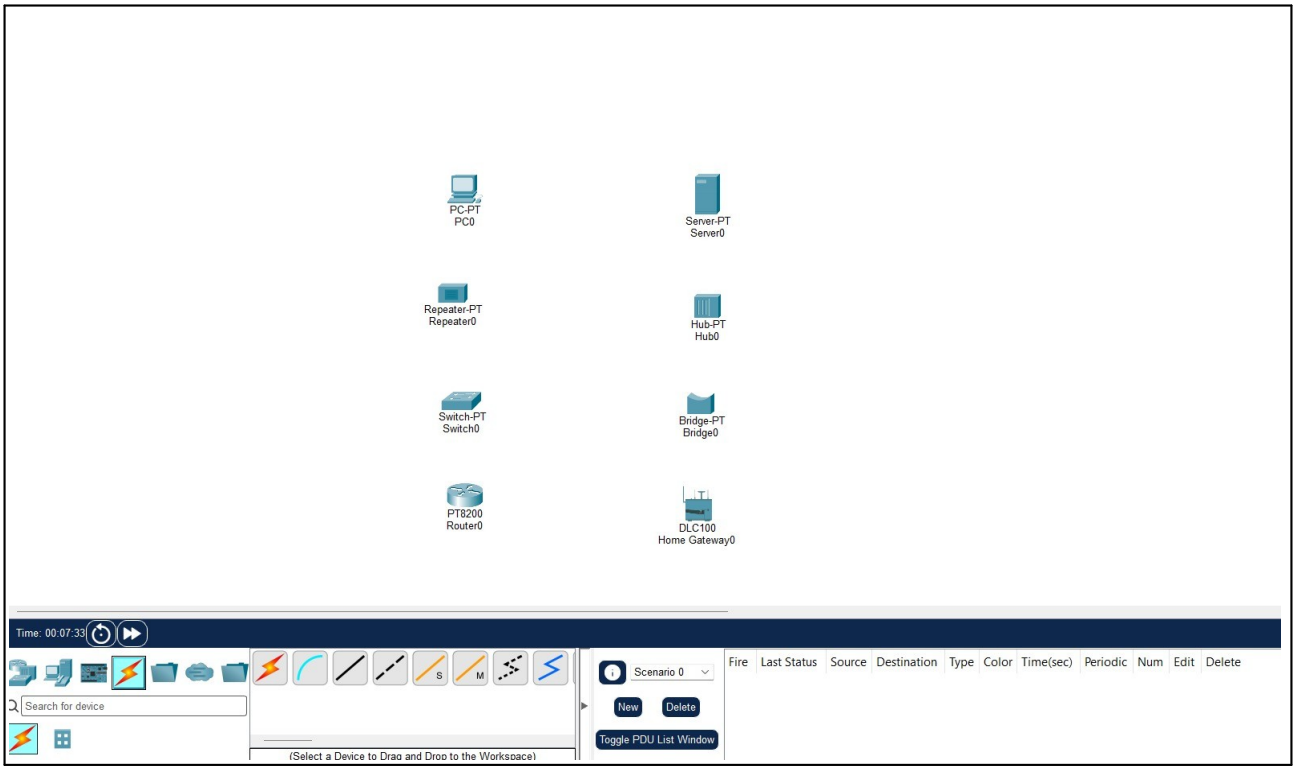


Fig: Network devices placed in Cisco Packet Tracer

Result

Thus the network components are studied in detail.